Project Documentation

HR Analytics and Employee Attrition Analysis

HR Analytics and Employee Attrition Analysis using Python, Pandas, Matplotlib, and Seaborn

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1. Introduction

Employee attrition is one of the major challenges faced by organizations. Understanding why employees leave can help companies improve retention strategies, reduce recruitment costs, and maintain workforce stability.

This project analyzes employee data to identify patterns and factors influencing attrition using data analytics techniques and visualization tools.

2. Problem Statement

Organizations often struggle to identify the reasons behind employee turnover. This project aims to analyze HR data and discover the factors that contribute to employee attrition.

3. Objectives

- Analyze employee attrition trends.
- Identify high-risk employee groups.
- Study the impact of income, satisfaction, experience, and overtime on attrition.
- Generate actionable insights through data visualization.

4. Tools and Technologies

Tool	Purpose
Python	Programming Language
Pandas	Data Cleaning and Analysis
NumPy	Numerical Operations
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Jupyter Notebook	Development Environment

5. Dataset Description

The dataset contains employee information including:

- Personal Details

- Employment Information
- Compensation Data
- Satisfaction Metrics
- Work-Life Balance Indicators
- Attrition Status

Dataset Size

- Records: 1470 Employees
- Features: 35 Columns

Data Source

- **Dataset Name:** IBM HR Analytics Employee Attrition & Performance Dataset
- **Source:** IBM Sample HR Analytics Dataset
- **Records:** 1,470 Employees
- **Features:** 39 Columns (after feature engineering and preprocessing)
- **Target Variable:** Attrition (Yes/No)
- The dataset contains employee demographics, compensation, work experience, satisfaction levels, performance metrics, and employment-related attributes used to analyze factors contributing to employee attrition.

6. Data Cleaning

The following cleaning activities were performed:

- Checked missing values.
- Checked duplicate records.
- Standardized column names.
- Verified data types.
- Identified constant-value columns.
- Removed unnecessary attributes.

7. Exploratory Data Analysis

Univariate Analysis

- Age Distribution
- Monthly Income Distribution
- Attrition Distribution

Bivariate Analysis

- Attrition by Gender
- Attrition by Department
- Attrition by Job Role
- Attrition by Overtime
- Job Satisfaction vs Attrition

Multivariate Analysis

- Correlation Analysis
- Heatmap Visualization

8. Visualizations

The project includes:

1. Attrition Distribution Chart
2. Age Distribution Histogram
3. Monthly Income Histogram
4. Attrition by Gender
5. Attrition by Department
6. Attrition by Job Role
7. Attrition by Overtime
8. Job Satisfaction Analysis
9. Correlation Heatmap

9. Key Findings

- Overtime is strongly associated with higher attrition.
- Employees with lower satisfaction levels are more likely to leave.
- Certain departments experience higher turnover rates.
- Income levels influence employee retention.
- Work-life balance significantly impacts attrition.

10. Conclusion

This HR Analytics project successfully explored employee attrition patterns using data analysis and visualization techniques. The findings reveal that factors such as overtime, job satisfaction, income, work-life balance, and job role play important roles in employee turnover.

The insights generated from this analysis can help HR teams make informed decisions, improve employee retention strategies, and create a more productive work environment. By leveraging data-driven approaches, organizations can proactively address attrition and enhance workforce stability.

11. Future Scope

- Attrition Prediction using Machine Learning.
- Employee Segmentation Analysis.
- Interactive Dashboard Development.
- HR KPI Monitoring Dashboard.

Author

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Data Analytics Project using Python, Pandas, Seaborn, Matplotlib, and Plotly

Dataset Type: Structured CSV Dataset

Source: IBM HR Analytics Employee Attrition & Performance Dataset

Domain: Human Resource Analytics